

HOUSTON HOOKS, TX, USA

WMO: 722429

Lat: 30.068N

Lon: 95.556W

Elev: 46

StdP: 100.77

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 53910

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	0.8	-8.5	1.8	5.5	-6.2	2.2	7.0	9.0	15.2	8.2	15.1	1.7	0	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	36.7	24.4	35.4	24.7	34.2	24.7	26.8	31.3	26.4	31.1	26.0	30.7	2.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.4	28.1	25.2	20.5	27.9	24.9	20.1	27.8	83.6	31.0	82.1	30.8	80.6	30.8	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	7.0	6.0	DB	-4.3	38.8	2.2	2.3	-5.9	40.5	-7.2	41.8	-8.5	43.1	-10.1	44.8	
			WB	-5.2	27.4	1.9	0.5	-6.5	27.8	-7.6	28.1	-8.7	28.4	-10.0	28.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	11.6	13.8	17.2	20.8	24.7	27.8	28.7	29.0	26.5	21.5	16.2	12.2
	DBStd	7.40	5.37	5.48	4.79	3.72	2.88	1.77	1.55	1.72	2.52	4.10	4.99	5.30
	HDD10.0	116	44	23	6	1	0	0	0	0	0	0	8	35
	HDD18.3	780	217	144	80	19	1	0	0	0	0	18	98	202
	CDD10.0	4084	94	129	228	324	457	533	579	589	497	357	193	104
	CDD18.3	1706	9	17	44	93	200	283	321	331	247	116	33	13
	CDH23.3	17330	29	71	237	652	1834	3123	3729	3893	2429	1059	221	53
	CDH26.7	7592	1	6	32	136	652	1436	1827	2004	1113	349	33	3
Wind	WSAvg	2.6	3.0	3.1	3.2	3.2	3.0	2.5	2.2	2.0	2.1	2.3	2.5	2.8
Precipitation	PrecAvg	1163	91	75	85	88	118	115	82	82	98	125	110	91
	PrecMax	1621	254	180	207	292	291	405	288	232	187	433	319	242
	PrecMin	431	11	9	14	6	0	0	0	0	23	0	15	25
	PrecStd	289	56	47	47	64	84	90	70	61	49	117	68	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	27.5	29.3	31.3	34.0	37.1	37.6	38.7	37.7	32.8	29.3	27.0
		MCWB	18.9	19.3	20.2	22.0	23.5	24.4	24.1	24.0	23.3	23.2	22.6	21.0
	2%	DB	23.9	25.6	27.4	29.4	32.5	35.1	36.2	36.9	35.0	31.7	27.5	24.8
		MCWB	18.5	19.2	20.2	21.4	23.5	24.4	24.6	24.6	23.7	23.5	21.1	20.3
	5%	DB	22.4	23.7	26.1	28.0	31.4	33.8	34.8	35.6	33.5	30.3	26.0	23.1
		MCWB	18.2	18.8	19.4	21.0	23.5	24.5	24.9	25.0	24.0	22.7	20.2	19.7
	10%	DB	21.0	22.2	24.3	27.2	30.3	32.7	33.7	34.1	32.3	28.7	24.1	21.7
		MCWB	17.7	18.3	18.7	20.4	23.2	24.6	25.0	25.1	23.9	21.7	19.9	18.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	21.7	22.4	24.3	25.8	26.9	27.1	27.3	26.9	26.0	24.0	22.7
		MCDB	23.1	24.3	26.3	28.3	30.6	31.8	31.4	31.4	30.4	29.3	26.8	24.7
	2%	WB	20.3	20.9	21.7	23.3	25.1	26.3	26.7	26.7	26.2	25.2	23.0	21.7
		MCDB	22.3	23.7	25.1	26.9	30.2	30.8	31.3	31.4	30.1	28.7	25.5	23.7
	5%	WB	19.4	20.2	21.0	22.7	24.6	25.8	26.2	26.3	25.7	24.5	22.1	20.8
		MCDB	21.7	22.7	24.1	26.1	29.2	30.4	31.1	31.3	29.9	28.0	24.7	22.9
	10%	WB	18.2	19.3	20.4	22.0	24.1	25.5	25.8	25.9	25.2	23.8	21.0	19.3
		MCDB	20.3	21.6	23.3	25.3	28.3	30.1	30.6	31.0	29.6	27.0	23.5	21.2
Mean Daily Temperature Range	5% DB	MDBR	11.1	10.8	11.1	10.8	10.2	9.8	10.1	10.9	10.9	12.1	11.7	11.1
		MCDBR	12.1	11.3	11.0	10.9	10.6	11.3	11.9	12.5	12.6	12.3	12.3	12.2
		MCWBR	7.4	6.4	5.2	4.7	3.8	3.0	3.0	3.1	3.9	5.0	6.5	8.2
	5% WB	MCDBR	9.9	9.4	9.4	8.6	9.1	9.1	9.9	10.6	9.8	9.9	10.1	10.6
	MCWBR	7.3	6.5	5.7	4.6	3.8	3.2	3.3	3.2	3.6	5.0	6.7	8.4	
Clear-Sky Solar Irradiance	taub	0.337	0.358	0.374	0.433	0.481	0.468	0.493	0.480	0.450	0.380	0.372	0.343	
	taud	2.498	2.451	2.401	2.237	2.140	2.231	2.182	2.221	2.267	2.451	2.416	2.488	
	Ebn at Noon	896	906	911	864	821	827	804	813	825	871	845	869	
	Edh at Noon	90	103	115	141	156	142	148	141	130	102	97	87	
All-Sky Solar Radiation	RadAvg	2.87	3.22	4.40	5.30	5.84	6.14	6.11	5.84	5.01	4.28	3.21	2.52	
	RadStd	0.28	0.58	0.39	0.36	0.43	0.50	0.53	0.44	0.50	0.54	0.33	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173	+133
(47)	Regional (37 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+109	+95

Nomenclature: See separate page